

- for any time-invariant random variables $Z_{i,t}$ such that $Z_{i,t} = Z_i$, double-demeaning annihilates it:
 $\ddot{Z}_{i,t} = 0$.

Proof. The zero-mean and zero-sum properties hold by definition of double-demeaning:

$$E[\ddot{X}_{i,t}] = E\left[X_{i,t} - E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T X_{i,s} + \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}]\right] \quad (33)$$

$$= E[X_{i,t}] - E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] + \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] = 0 \quad (34)$$

and

$$\sum_{t=0}^T \ddot{X}_{i,t} = \sum_{t=0}^T X_{i,t} - \sum_{t=0}^T E[X_{i,t}] - \sum_{t=0}^T \frac{1}{T+1} \sum_{s=0}^T X_{i,s} + \sum_{t=0}^T \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}] \quad (35)$$

$$= \sum_{t=0}^T X_{i,t} - \sum_{t=0}^T E[X_{i,t}] - \sum_{s=0}^T X_{i,s} + \sum_{s=0}^T E[X_{i,s}] = 0 \quad (36)$$

For the idempotent property, first note that by definition of double-demeaning, rearranging terms we can write $E[\ddot{X}_{i,t} \ddot{Z}_{i,t}]$ as

$$= E\left[\left(X_{i,t} - \frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \ddot{Z}_{i,t}\right] - E\left[\left(E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}]\right) \ddot{Z}_{i,t}\right] \quad (37)$$

$$= E\left[\left(X_{i,t} - \frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \ddot{Z}_{i,t}\right] - \left(E[X_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T E[X_{i,s}]\right) \cdot E[\ddot{Z}_{i,t}] \quad (38)$$

By the zero-mean property $E[\ddot{Z}_{i,t}] = 0$, the second term of the above expression is zero. Summing the first term over t , we have

$$\sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] - \sum_{t=0}^T E\left[\left(\frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \ddot{Z}_{i,t}\right] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] - E\left[\left(\frac{1}{T+1} \sum_{s=0}^T X_{i,s}\right) \sum_{t=0}^T \ddot{Z}_{i,t}\right] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}] \quad (39)$$

where the last equality follows from the zero-sum property $\sum_{t=0}^T \ddot{Z}_{i,t} = 0$. This proves $\sum_{t=0}^T E[\ddot{X}_{i,t} \ddot{Z}_{i,t}] = \sum_{t=0}^T E[X_{i,t} \ddot{Z}_{i,t}]$. Similarly we can show $\sum_{t=0}^T E[\ddot{X}_{i,t} \ddot{Z}_{i,t}] = \sum_{t=0}^T E[\ddot{X}_{i,t} Z_{i,t}]$.

The annihilating property of double-meaning holds by definition for any time-invariant random variable. Specifically, plugging in $Z_{i,t} = Z_i$ we have

$$\ddot{Z}_{i,t} = Z_{i,t} - E[Z_{i,t}] - \frac{1}{T+1} \sum_{s=0}^T Z_{i,s} + \frac{1}{T+1} \sum_{s=0}^T E[Z_{i,s}] \quad (40)$$

$$= Z_i - E[Z_i] - Z_i + E[Z_i] = 0 \quad (41)$$

□

A.1 Expression of the weight

In this section, we provide detailed definitions for the three elements in the expression of the weight $\omega_{e,\ell}^\ell = \sigma_{e,\cdot}' \mathbf{\Delta}_{e+\ell'} A_\ell^{-1}$. Recall $\omega_{e,\ell}^\ell$ is the weight associated with $CATT_{e,\ell'}$ in the linear combination underlying μ_ℓ :

$$\begin{aligned}
\omega_{e,\ell'}^\ell &= \mathbf{e}_\ell^\top \underbrace{\left(\sum_{t=0}^T E[\mathbf{\ddot{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1}}_{:= (A_\ell^{-1})^\top} E[\mathbf{\ddot{D}}_{i,t} D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\}] \\
&= (A_\ell^{-1})^\top \sum_{t=0}^T Cov(\mathbf{D}_{i,t} - \frac{1}{T+1} \sum_{t=0}^T \mathbf{D}_{i,t}, D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\} - \frac{1}{T+1} \sum_{t=0}^T D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\}) \\
&= (A_\ell^{-1})^\top Pr\{E_i = e\} \cdot \left((E[\mathbf{D}_{i,e+\ell'} - \frac{1}{T+1} \sum_{t=0}^T \mathbf{D}_{i,e+\ell'} | E_i = e]) - (E[\mathbf{D}_{i,e+\ell'}] - E[\frac{1}{T+1} \sum_{t=0}^T \mathbf{D}_{i,e+\ell'}]) \right) \\
&= (A_\ell^{-1})^\top \sum_{e'} E[\mathbf{D}_{i,e+\ell'} - \frac{1}{T+1} \sum_t \mathbf{D}_{i,t} | E_i = e'] \cdot Cov(\mathbf{1}\{E_i = e\}, \mathbf{1}\{E_i = e'\}) \\
&:= (\sigma_{e,\cdot})^\top \mathbf{\Delta}_{e+\ell'} A_\ell^{-1}
\end{aligned}$$

We note that

- The first term is the vector $\sigma_{e,\cdot} = (\sigma_{e,e'})_{e' \in \text{supp}(E_i)}$ of the covariance between cohort indicators. Each entry $\sigma_{e,e'} = Cov(\mathbf{1}\{E_i = e\}, \mathbf{1}\{E_i = e'\})$ is the covariance between being in cohort e and being in cohort e' .
- The second term is the matrix $\mathbf{\Delta}_{e+\ell'}$ of demeaned relative time indicators. The rows of the matrix are indexed by cohorts $e' \in \text{supp}(E_i)$ and the columns of the matrix are indexed by relative time indicators included in the specification (7). The entry that corresponds to cohort e' and relative time indicator $D_{i,t}^\ell$ is $E[D_{i,e+\ell'}^\ell - \frac{1}{T+1} \sum_t D_{i,t}^\ell | E_i = e']$. Note that the sum $\sum_t D_{i,t}^\ell$ is equal to $\mathbf{1}\{E_i \in \mathcal{I}_\ell\}$ the indicator for whether this individual ever experience the relative time ℓ .
- The third term is the row vector of A^{-1} that corresponds to the relative time indicator $D_{i,t}^\ell$. Here A is the covariance matrix of demeaned relative time indicators included in the specification (7). Specifically, the entry that corresponds to the covariance between demeaned $D_{i,t}^\ell$ and $D_{i,t}^{\ell'}$ is

$$A = \sum_t Cov\left(D_{i,t}^\ell - \frac{1}{T+1} \sum_t D_{i,t}^\ell, D_{i,t}^{\ell'} - \frac{1}{T+1} \sum_t D_{i,t}^{\ell'}\right).$$

Plugging in $\sum_t D_{i,t}^\ell = \mathbf{1}\{E_i \in \mathcal{I}_\ell\}$ gives the expression in the main text.

B Proofs of decomposition results

Proof of Proposition 1

Proof. Collect the relative time indicators in a column vector $\mathbf{D}_{i,t} = (\mathbf{1}\{t - E_i \in g\})_{g \in \mathcal{G}}^\top$. Collect their corresponding coefficients in a column vector $\mu = (\mu_g)_{g \in \mathcal{G}}^\top$. Partialling out the unit and time fixed effects, Regression (4) is

$$\ddot{Y}_{i,t} = \mu^\top \ddot{\mathbf{D}}_{i,t} + v_{i,t} \quad (42)$$

where $\ddot{X}_{i,t}$ is time- and cross-sectional demeaned version of $X_{i,t}$ as defined in Definition 3. By the idempotent property of Lemma 1

$$\mu_g = \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,t}] \quad (43)$$

where $\mathbf{e}_g$ is a column vector with one in the entry corresponding to the entry of $\mathbf{1}\{t - E_i \in g\}$ in $\mathbf{D}_{i,t}$, and zero otherwise.

To further develop the expression for the population regression coefficients μ_g , we note that by Lemma 1, we have $\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,0}^\infty] = \sum_{t=0}^T E[\mathbf{D}_{i,t} \ddot{Y}_{i,0}^\infty] = 0$ since $Y_{i,0}^\infty$ is time-invariant, and $E[\ddot{\mathbf{D}}_{i,t}] = 0$ by the zero-mean property.

$$\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,t}] = \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,t}] - \underbrace{\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} Y_{i,0}^\infty]}_{=0} = \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} (Y_{i,t} - Y_{i,0}^\infty)] = \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} E[Y_{i,t} - Y_{i,0}^\infty | E_i]] \quad (44)$$

$$= \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} E[Y_{i,t} - Y_{i,0}^\infty | E_i]] - \underbrace{\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t}] E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (45)$$

$$= \sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} (E[Y_{i,t} - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty])]. \quad (46)$$

We can abbreviate the term in the parentheses as $f(E_i, t)$ to emphasize it is a function of E_i and t . Since E_i and t take on discrete values, we can write

$$E[Y_{i,t} - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty] =: f(E_i, t) = \sum_{e=0}^{\infty} f(e, t) \cdot \mathbf{1}\{E_i = e\} \quad (47)$$

$$= \sum_{e=0}^{\infty} \sum_{\ell=-T}^T f(e, e + \ell) \cdot \mathbf{1}\{t - e = \ell\} \cdot \mathbf{1}\{E_i = e\} \quad (48)$$

$$= \sum_{\ell=-T}^T \sum_{e=0}^{\infty} D_{i,t}^\ell \mathbf{1}\{E_i = e\} \cdot f(e, e + \ell). \quad (49)$$

Even though we sum over $e \in \{0, \dots, \infty\}$, if there are no never-treated units (or no units in a particular cohort

e'), then we have no units take on values $E_i = \infty$ (or $E_i = e'$). The range of the summation is therefore still sensible. Note that we replace $\mathbf{1}\{t - e = \ell\} \cdot \mathbf{1}\{E_i = e\}$ with $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ because

$$D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\} = \mathbf{1}\{t - E_i = \ell\} \cdot \mathbf{1}\{E_i = e\} = \mathbf{1}\{t - e = \ell\} \cdot \mathbf{1}\{E_i = e\}. \quad (50)$$

Using the above expression, the coefficient μ_g can be written as

$$\mu_g = \mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} \sum_{t=0}^T \sum_{\ell=-T}^T \sum_{e=0}^\infty E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}] f(e, e + \ell) \quad (51)$$

$$= \sum_{t=0}^T \sum_{l \in g} \sum_{e=0}^\infty \underbrace{\mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}]}_{=\omega_{e,\ell}^g} f(e, e + \ell) \quad (52)$$

$$+ \sum_{t=0}^T \sum_{g' \neq g} \sum_{\ell \in g'} \sum_{e=0}^\infty \underbrace{\mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}]}_{=\omega_{e,\ell}^g} f(e, e + \ell) \quad (53)$$

$$+ \sum_{t=0}^T \sum_{\ell \in g^{excl}} \sum_{e=0}^\infty \underbrace{\mathbf{e}_g^\top \left(\sum_{t=0}^T E[\ddot{\mathbf{D}}_{i,t} \mathbf{D}_{i,t}^\top] \right)^{-1} E[\ddot{\mathbf{D}}_{i,t} D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}]}_{=\omega_{e,\ell}^g} f(e, e + \ell). \quad (54)$$

The superscript g in $\omega_{e,\ell}^g$ indexes μ_g . The subscript ℓ in $\omega_{e,\ell}^g$ indexes $f(e, e + \ell)$. The expression above the braces makes it clear that the weight $\omega_{e,\ell}^g$ is equal to the population regression coefficient on $\mathbf{1}\{t - E_i \in g\}$ from regressing $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all the bin indicators i.e. $\{\mathbf{1}\{t - E_i \in g\}\}_{g \in \mathcal{G}}$ and two-way fixed effects included in (4). This proves Proposition 1. We next show several properties of these weights in the expression of μ_g :

1. For relative times of μ_g 's own bin i.e. $\ell \in g$, the weights sum to one $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$.
2. For relative times belonging to some other bin i.e. $\ell \in g'$ and $g' \neq g$, the weights sum to zero $\sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g = 0$ for each bin g' .
3. For relative times not contained in $\mathcal{G}$ i.e. $\ell \in g^{excl}$, the weights sum to negative one $\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = -1$.
4. If there are never-treated units i.e. $\infty \in \text{supp}(E_i)$, we have $\omega_{\infty,\ell}^g = 0$ for all g and ℓ .